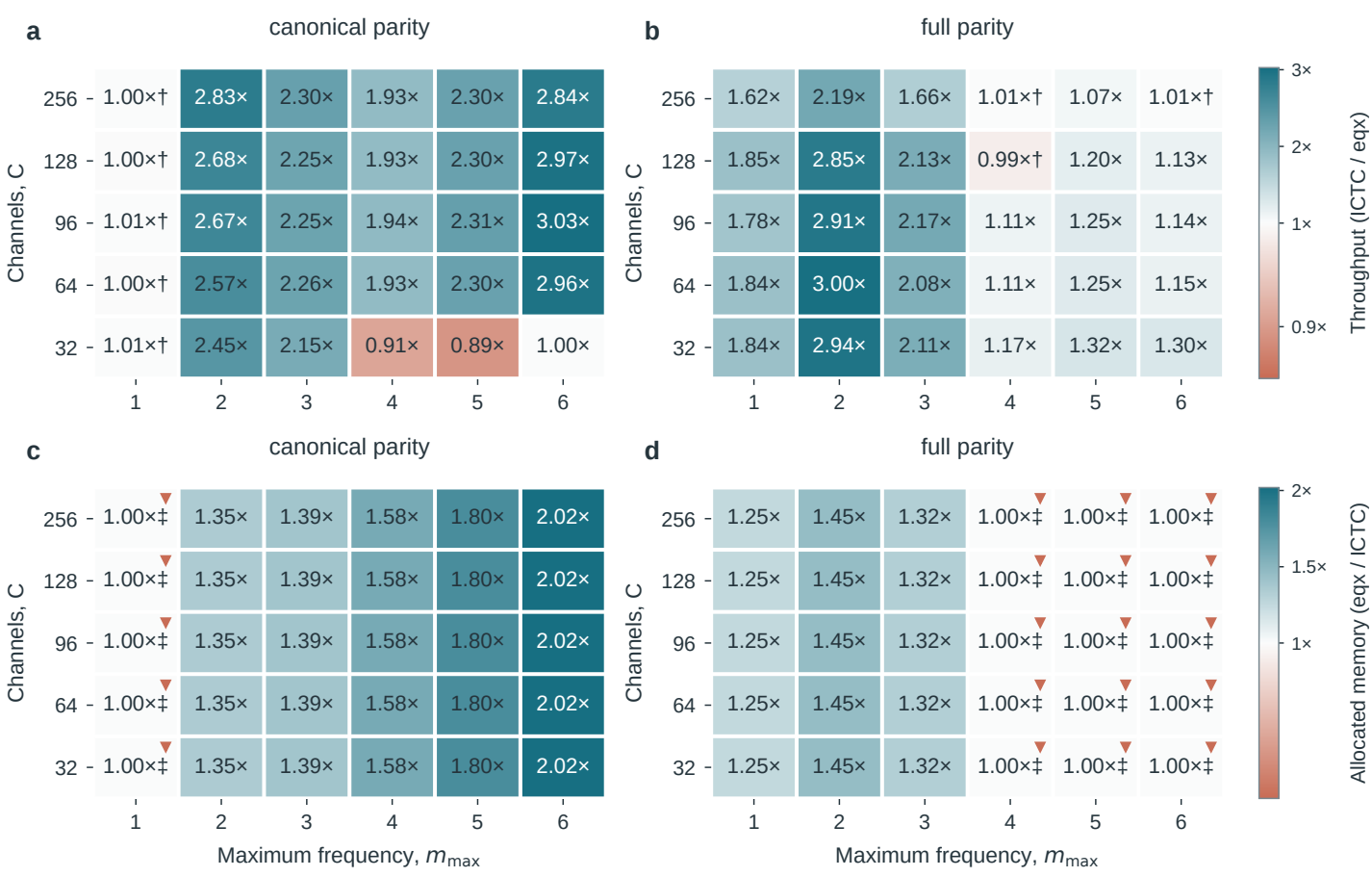

ICTC-O2 HP vs eqx: throughput and peak memory
O(2) tensor products · Forward · N = 131,072 · C = 32–256 · m_max = 1–6
a
canonical parity
b
full parity
c
canonical parity
d
full parity
Channels, C
256
128
96
64
32
1 2 3 4 5 6
Maximum frequency, m_{max}
a: 1.00×† 2.83× 2.30× 1.93× 2.30× 2.84 / 1.00×† 2.68× 2.25× 1.93× 2.30× 2.97 / 1.01×† 2.67× 2.25× 1.94× 2.31× 3.03 / 1.00×† 2.57× 2.26× 1.93× 2.30× 2.96 / 1.01×† 2.45× 2.15× 0.91× 0.89× 1.00×
b: 1.62× 2.19× 1.66× 1.01×† 1.07× 1.01×† / 1.85× 2.85× 2.13× 0.99×† 1.20× 1.13× / 1.78× 2.91× 2.17× 1.11× 1.25× 1.14× / 1.84× 3.00× 2.08× 1.11× 1.25× 1.15× / 1.84× 2.94× 2.11× 1.17× 1.32× 1.30×
c: 1.00×‡ 1.35× 1.39× 1.58× 1.80× 2.02×
d: 1.25× 1.45× 1.32× 1.00×‡ 1.00×‡ 1.00×‡
Throughput (ICTC / eqx)
3×
2×
1×
0.9×
Allocated memory (eqx / ICTC)
2×
1.5×
1×
† IQR overlap * isolated timing ‡ memory increase
v0.1.5